



Daily Drilling Report

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 5

Report For 01-Jul-21

Operator:	University of Utah	Rig:	Frontier Rig 16	Spud Date:	28-Jun-21	Daily Cost / Mud (\$):	---
Measured Depth (ft):	2699	Last Casing:	16.000 at 416	Wellbore:	Original Wellbore	AFE No.	---
Vertical Depth (ft):	2699	Next Casing:	11.750 at 3,300	RKB Elevation (ft):	30.40	AFE (\$)	Actual (\$)
Proposed TD (ft):	9500	Last BOP Test:	29-Jun-21	Job Reference RKB (ft):	---	---	---
Hole Made (ft) / Hrs:	310 / 22.75	Next BOP Test:	19-Jul-21	Working Interest:	---	Totals:	---
Average ROP (ft/hr):	13.63	LOT (lbs/gal):	0.58	Well Cost (\$):	---	---	---
Drilling Days (act./plan):	5/21	Flat Days (act./plan):	0/9	Total Days (act./plan):	5/30	Days On Location:	5
Pers/Hrs: Operator:	3 / 36	Contractor:	14 / 148	Service:	4 / 48	Other:	3 / 36
				Total:	24 / 268		

Safety Summary: No incidents or events reported. Conducted Crown Check, Safety Meeting.

Current Operations: Cleaned and cleared rig floor. Rigged up and ran 11 3/4" casing to 2,800' (report time)

Planned Operations:

- Run 11 3/4" casing to set depth.
- Rig down Casing crew.
- Run in the hole with stab-in tool.
- Circulate and cement 11 3/4" casing.
- Nipple down 20" BOPE.

Toolpusher: Steve Caldwell , Justin Bristol

Wellsite Supervisors: Virgil Welch , Brian Gresham

Tel No.: 7132807438

Operations Summary

From	To	Elapsed	End MD(ft)	Code	Operations Description	Non-Prod
0:00	2:00	2.00	2,699	3-6-2	Pulled out of the hole from 951' to Surface. Broke out bit and inspected MWD. No visible wear on roller reamer or stabs. Motor had minimal movement in bearing assembly. cleaned and cleared rig floor.	
2:00	4:00	2.00	2,699	3-6-1	Make up 14 3/4" Bit and trip in the hole with BHA to 951'.	
4:00	5:15	1.25	2,699	10-6-3	Trip in the hole from 951' to 2,699'.	
5:15	15:30	10.25	3,009	3-2-3	Drilled 14-3/4" hole from 2,699' to 3,009' with full returns.	
15:30	16:00	0.50	3,009	3-2-3	After making a hook ROP dropped from 15'/hour to 0'/hour. Perform step test from 5K to 30K no ROP / No change in diff pressure and no pressure loss and no hook load change but found excessive amounts of metal on ditch magnet mix dry job.	
16:00	19:30	3.50	3,009	3-6-2	Pulled out of the hole from 3,009' to 951' with no issues. hoe taking proper fill. Stopped and removed rotating rubber at 1,100'.	
19:30	0:00	4.50	3,009	3-6-2	Racked back HWDP. Racked back 8" drill collars. Laid out Cubic sensor, Diffuser sub, along with float sub. Pulled up and broke out bit. laid out the remaining RSS tools. (Note: Bit was 3.75" under gage, 3.5" of the center of the bit was washed out with 4 jets missing. RSS had signs of wear below the pads approximately .25" wide by .25" deep. No signs of wear on the stabilizers or roller reamer. Motor had approximately .25" of play in bearing assembly.	

Management Summary

Pulled out of the hole from 951' to Surface.
Picked up 14 3/4" PDC bit.
Tripped in the hole to 2,699'.
Drilled 14-3/4" hole from 2,699' to 3,009' with full returns. Decision was made to pull out of the hole at 3,009' due to low ROP.
Pulled out of the hole from 3,009' to surface.
Laid out RSS assembly. cleaned and cleared rig floor for casing run.

Comments

Bit was 3.75" under gage with 3.5" of the center of the bit was washed out with 4 jets missing. RSS had signs of wear below the pads approximately 0.25" wide by 0.25" deep. No signs of wear on the stabilizers or roller reamer.

Casing/Tubular Information

Type	Size (ins)	Top MD (ft)	Top TVD (ft)	Bottom MD (ft)	Bottom TVD (ft)	Hole Section	OH Diam. (ins)	Nom. Wgt. (lbs)	Nominal Grade	LOT (lbs/gal)
FULL	16.000	0		416		SURF	22.000	65	K-55	0.58

Mud Information

%																Gels			Temp		Mud
Dens.	Vis	PV	YP	Filt.	Cake	pH/ES	Solids	Oil	Water	Sand	LGS	Cl	Ca	CaCl	10s	10m	30m	In	Out	Loss	
01-Jul-21 20:00 at Depth 3,009 ft Mud Pits, Type: Low Solids Non-Dispersed																					
9.00	42	14	15	7.8	1	11	9	0	92	1	3.2	800	120		5	9	16	116	109		

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Mud Consumables

Item Description	Qty.	Cost	Item Description	Qty.	Cost
Barite - 100#SK	20	---	Engineering - EACH	1	---
Gel - 100#SK	96	---	perdiem - OTHER	1	---
Zanflow - 25#SK	5	---			

Bit/BHA/Workstring Information

				Depth		This Run		R.O.P.				Mud		Pump						
No	Run	Make	Model	Diam	In	Dist	Hrs	Avg	Max	WOB	RPM	Torque	Wt	Flow	Press	J. Vel	P. Drp	HHP	JIF	
3	1	NOV	TKC66	14.750	433	2,266	9	251.8								267	572	301	1118	
Jets: 15 15 15 15 16 16					Out: 2699		Grade: Cutter:		3 / 1		Dull: LT / CT		Wear: N		Brgs: X		Gge:		Pull: PR	
4	1	NOV	TKC63	14.750	2699	310	10.25	30.2	75.0	40	188	15	9	900	2560	267	572	301	1118	
Jets: 15 15 15 15 16 16					Out: 3009		Grade: Cutter:		4 / 8		Dull: RG / CR		Wear: A		Brgs: X		Gge: 3.75		Pull: PR	
BHA - No. 4 - BIT, OTHER, STAB, MWD, 4 OTHER, MMTR, RR, FLOAT, 2 OTHER, DC, XO, HWDP = 956.61																				

Drilling Parameters

Depth (ft)		ROP (ft/hr)		WOB (lbs)		RPM		Torque (ft lbs)		Flow (gals/min)		Pressure (psi)	
From	To	Avg	Max	Avg	Max	Avg	Max	Avg	Max	Avg	Max		
2,699	3,009	30.2	80.0	40	70	188	220	15	21	900	950	2,587	
Annular Velocity: Drill Collars:				199.3	Drill Pipe:				129.3				

Miscellaneous Drilling Parameters

Hook Loads (lbs):		Off Bottom Rotate:		145	Pick Up:		165	Slack Off:		135	Drag Avg/Max:		10 / 15
Slow Circulation Data:													
Pump 1:		30 spm	95 psi	45 spm	210 psi	60 spm	350 psi						
Pump 2:		30 spm	95 psi	45 spm	210 psi	60 spm	350 psi						
Pump 3:		30 spm	95 psi	45 spm	210 psi	60 spm	350 psi						
Hours on BHA:		Since Inspection:		19.25	Total:		19.25	Jars:		0			
Hours on Casing/Liner:		Rotating:		19.25 / 10.25		Tripping:		11 / 8		☐ Wear Bushing Installed			

Mud Log Information

Depth (ft)		TVD (ft)		Gas (Units)		Gas		Drilling		Pore		Mud		Shale		ROP	
From	To	From	To	Avg	Max	at Depth	Connect.	Trip	Exp.	Press	Dens.	Dens.	Dens.	Shale	Sand		
2,650	2,670	2,650	2,670	0	0	0	0	0	0		9.00						
Formation Name: 1050% Rhyolite 50-90% Granite																	
Lithology:																	
2,670	2,700	2,670	2,700	0	0	0	0	0	0		9.00						
Formation Name:																	
Lithology: 100% Granite																	
2,700	3,000	2,700	3,000	0	0	0	0	0	0		9.00						
Formation Name:																	
Lithology: 100% Granite																	

Rig Information

Equipment Problems:	Generators having issues with over heating with higher pump rates.
Location Condition:	Good
Transport:	

Solids Control Information

Screen Sizes:	Top	Middle 1	Middle 2	Bottom	Equipment Usage (Hrs):	
Shaker No 1:	60	60	60	60		
Shaker No 2:	60	60	60	60	Centrifuge 1: 14 (Solids Removal)	Centrifuge 2: 14
Shaker No 3:	60	60	60	60		

Bulk Inventory

Item Type	Units	Beginning	Used	Received	On Hand	Net
Rig Fuel		8,096	2,844		5,252	



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Drill Pipe Inventory

DP Size	Joints	Weight	Grade	Thread	DP Size	Joints	Weight	Grade	Thread
5"	114	19.5	S-135	4.5IF					

Safety Information

Meetings/Drills	Time	Description
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Safety 60 Two 30 min Pre tour safety meetings with both crews. Handling BHA, Tripping pipe, Rigging up and running casing.

First Aid Treatments: 0 Medical Treatments: 0 Lost Time Incidents: 0 Days Since LTI:

☐ BOP Test ☒ Crownamatic Check

Weather Information

Sky Condition:	Clear	Visibility:	10
Air Temperature:	70 degF	Bar. Pressure:	29.83
Wind Speed/Dir:	6 / NNW	Wind Gusts:	1